```
import pandas as pd
import numpy as np
df=pd.read_csv('/content/titanic_data.csv')
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

Next steps: Generate code with df New interactive sheet

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, f1_score
from sklearn.preprocessing import LabelEncoder
```

```
df=df[['Survived','Pclass','Sex','Age','SibSp','Parch','Ticket','Fare','Cabin','Embarked']]
df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	0	3	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	1	1	female	38.0	1	0	PC 17599	71.2833	C85	C
2	1	3	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	1	1	female	35.0	1	0	113803	53.1000	C123	S
4	0	3	male	35.0	0	0	373450	8.0500	NaN	S

Next steps: Generate code with df New interactive sheet

```
df.isnull().sum()
```

	0
Survived	0
Pclass	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df['Age'].fillna(df['Age'].median(),inplace=True)
```

```
/tmp/ipykernel_156/1527141296.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through ch
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Age'].fillna(df['Age'].median(),inplace=True)
```

```
df.drop(columns=['Cabin'],inplace=True)
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0],inplace=True)
```

```
/tmp/ipykernel_156/3717664479.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through ch
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df['Embarked'].fillna(df['Embarked'].mode()[0],inplace=True)
```

```
df.isnull().sum()
```

	0
Survived	0
Pclass	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 9 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Survived  891 non-null    int64
 1   Pclass    891 non-null    int64
 2   Sex       891 non-null    object
 3   Age       891 non-null    float64
 4   SibSp     891 non-null    int64
 5   Parch     891 non-null    int64
 6   Ticket    891 non-null    object
 7   Fare      891 non-null    float64
 8   Embarked  891 non-null    object
dtypes: float64(2), int64(4), object(3)
memory usage: 62.8+ KB
```

```
le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])
df['Ticket'] = le.fit_transform(df['Ticket'])
df['Embarked'] = le.fit_transform(df['Embarked'])
```

```
df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	0	3	1	22.0	1	0	523	7.2500	2
1	1	1	0	38.0	1	0	596	71.2833	0
2	1	3	0	26.0	0	0	669	7.9250	2
3	1	1	0	35.0	1	0	49	53.1000	2
4	0	3	1	35.0	0	0	472	8.0500	2

Next steps: Generate code with df New interactive sheet

```
x=df.drop('Survived', axis=1)
y=df['Survived']
```

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

```
depth_values = [2,4,6,8]
criteria = ['gini', 'entropy']

result = []
for c in criteria:
    for d in depth_values:

      model = DecisionTreeClassifier(
          criterion=c,
          max_depth=d,
          random_state=42)
      model.fit(x_train, y_train)
      y_pred = model.predict(x_test)
      accuracy = accuracy_score(y_test, y_pred)
      f1 = f1_score(y_test, y_pred)
      cm = confusion_matrix(y_test, y_pred)

      print("\n==============================")
      print(f"Criterion:", c)
      print(f"Max Depth:", d)

      print("Accuracy:", accuracy)

      print("f1 score:", f1)

      print("Confusion Matrix")
      print(cm)

      result.append([c, d, accuracy, f1])
```

```
f1 score: 0.75
Confusion Matrix
[[89 16]
 [20 54]]

==============================
Criterion: gini
Max Depth: 6
Accuracy: 0.8324022346368715
f1 score: 0.782608695652174
Confusion Matrix
[[95 10]
 [20 54]]

==============================
Criterion: gini
Max Depth: 8
Accuracy: 0.7597765363128491
f1 score: 0.6861313868613139
Confusion Matrix
[[89 16]
 [27 47]]

==============================
Criterion: entropy
Max Depth: 2
Accuracy: 0.7653631284916201
f1 score: 0.6379310344827587
Confusion Matrix
[[100   5]
 [ 37  37]]

==============================
Criterion: entropy
Max Depth: 4
Accuracy: 0.7988826815642458
f1 score: 0.75
Confusion Matrix
[[89 16]
 [20 54]]

==============================
Criterion: entropy
```

```
results_df=pd.DataFrame(result, columns=['Criterion', 'Max Depth', 'Accuracy', 'f1 score'])
results_df
```

1 to 8 of 8 entries Filter

index	Criterion	Max Depth	Accuracy	f1 score
0	gini	2	0.7653631284916201	0.6379310344827587
1	gini	4	0.7988826815642458	0.75
2	gini	6	0.8324022346368715	0.782608695652174
3	gini	8	0.7597765363128491	0.6861313868613139
4	entropy	2	0.7653631284916201	0.6379310344827587
5	entropy	4	0.7988826815642458	0.75
6	entropy	6	0.776536312849162	0.6721311475409836
7	entropy	8	0.7206703910614525	0.6268656716417911

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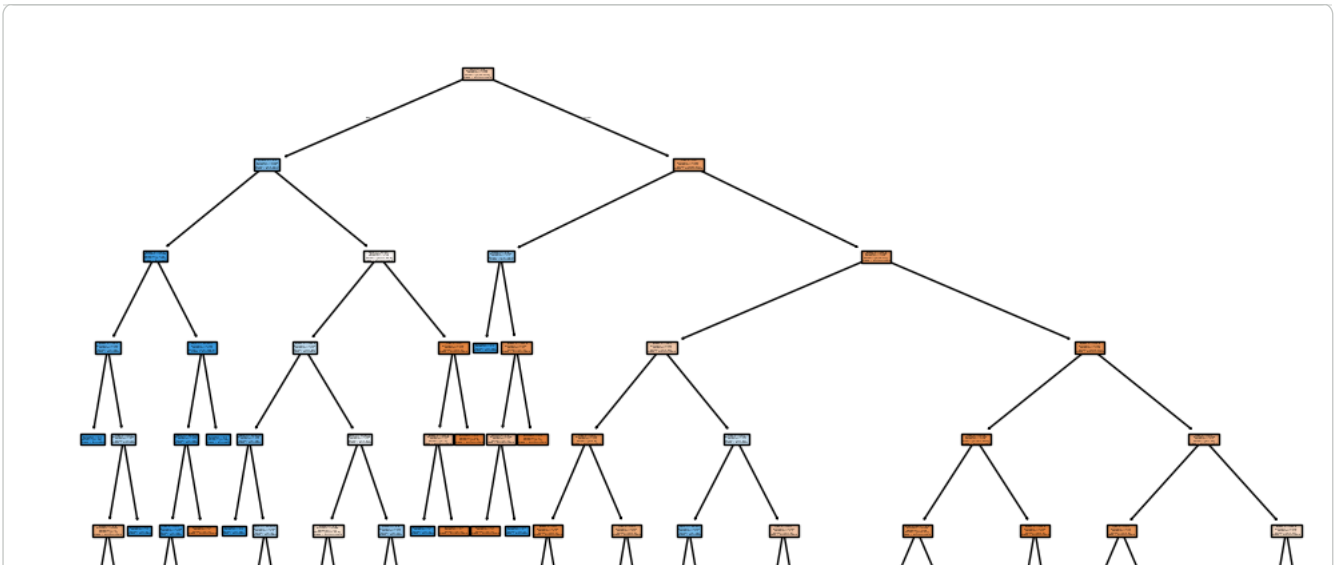
Next steps: Generate code with results_df New interactive sheet

```
results_df.sort_values(by='f1 score', ascending=False)
```

	Criterion	Max Depth	Accuracy	f1 score
2	gini	6	0.832402	0.782609
1	gini	4	0.798883	0.750000
5	entropy	4	0.798883	0.750000
3	gini	8	0.759777	0.686131
6	entropy	6	0.776536	0.672131
0	gini	2	0.765363	0.637931
4	entropy	2	0.765363	0.637931
7	entropy	8	0.720670	0.626866

```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 8))
plot_tree(model, feature_names=x.columns,
        class_names=['Not Survived', 'Survived'],
          filled=True)
plt.show()
```